



Automation Frameworks – Keyword-Driven & Data-Driven (Basic Idea):

Introduction or Definition

An **automation testing framework** is a structured combination of test tools and practices designed to automate test cases efficiently and consistently.

It defines how test scripts are created, organized, and executed.

Two commonly used frameworks:

- **Keyword-Driven Framework**
- **Data-Driven Framework**

These frameworks improve **reusability**, **maintainability**, and **scalability** of test scripts.

Keyword-Driven Framework

Definition:

A **keyword-driven framework** uses **predefined keywords** to represent test actions. Testers write tests using these keywords instead of traditional code.

Key Points:

- Test logic is separated from the code.
- Test cases are written in **Excel/CSV/XML**, using action keywords like click, input, select.
- Framework maps each keyword to an actual function.

Example:

Keyword	Object	Value
click	loginBtn	
input	username	user123

Keyword	Object	Value
input	password	pass123
click	submitBtn	

Mapping Code (in Python or Java):

```

if keyword == "click":
    click(object)

elif keyword == "input":
    type_text(object, value)

```

Data-Driven Framework

Definition:

A **data-driven framework** separates test data from test scripts. Test scripts run multiple times with different input values stored in files (Excel, CSV, JSON).

Key Points:

- Focuses on **testing the same logic with various datasets**.
- Reduces code duplication.
- Suitable for **form validations, boundary testing, login scenarios**, etc.

Example:

Test Data (Excel):

username	password
user1	pass123
user2	pass456

Single Script Logic:

```

for row in test_data:
    login(row["username"], row["password"])

```

Framework Comparison Table

Feature	Keyword-Driven	Data-Driven
Focus	Actions (keywords)	Input data
Test Case Format	Excel/XML with keywords	Excel/CSV with values
Suitable For	Non-coders, reusable logic	Multiple data scenarios
Test Logic Location	In keyword functions	In test script

Tools Supporting These Frameworks

- **Selenium** (combined with Excel/TestNG)
 - **Robot Framework** (Keyword-Driven)
 - **JUnit/TestNG** (with DataProvider for Data-Driven)
 - **Katalon Studio, UFT/QTP**
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MCQ Questions with Answers and Explanations

Theory Questions

1. What is the main purpose of an automation framework?

- a) To replace manual testers
- b) To organize and standardize test scripts
- c) To write programming languages
- d) To design the UI

Answer: b — Frameworks provide structure and reusability for tests.

2. In a keyword-driven framework, test cases are written using:

- a) Code only
- b) Random values
- c) Predefined action words
- d) SQL commands

Answer: c — Keywords like click, enter, verify are used.

3. A data-driven framework is best when:

- a) Only one test is needed
- b) The app has no input fields
- c) Multiple data sets need to be tested
- d) Tests are run without tools

Answer: c — Data-driven frameworks test the same logic with many input values.

4. Which format is commonly used for input in keyword-driven tests?

- a) MP3
- b) HTML
- c) Excel/CSV/XML
- d) ZIP

Answer: c — Keyword tables are maintained in Excel, CSV, or XML files.

5. Which tool best supports both keyword and data-driven testing?

- a) Windows Notepad
- b) Selenium
- c) Instagram
- d) Paint

Answer: b — Selenium supports both frameworks with custom implementation.

Coding/Scenario Questions

6. You are testing a login page with 100 username-password pairs. Which framework is most efficient?

- a) Linear scripting
- b) Data-driven framework
- c) Keyword-driven
- d) Record-playback

Answer: b — Data-driven tests handle multiple inputs easily.

7. A test case row contains: input, emailField, test@example.com. This is an example of:

- a) SQL injection
- b) Data-driven command
- c) Keyword-driven step
- d) Manual test case

Answer: c — It's a keyword-driven action using input.

8. Which keyword could be used in a keyword-driven framework?

- a) CREATE TABLE
- b) EnterValue
- c) click
- d) root

Answer: c — click is a common keyword in such frameworks.

9. You write test data in Excel and run the same script 5 times with different inputs. This is:

- a) Static testing
- b) Unit testing
- c) Keyword-driven
- d) Data-driven

Answer: d — Using multiple sets of data in one script = Data-driven.

10. In a data-driven framework using JUnit/TestNG, which feature helps in supplying data?

- a) @BeforeClass
- b) @AfterSuite
- c) @DataProvider
- d) @Ignore

Answer: c — @DataProvider is used for feeding multiple data sets.